Feynman integrals: from Integration-by-Parts identities to Twisted Cohomology

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Abstract

Notes for a 2-hours seminar given for the *Research seminar* module in the Master by Research programme at the University of Hertfordshire, England.

Contents

1 Feynman integrals

In these two seminars we introduce the study of Feynman Integrals, the modern approach to their computation (Integration-By-Parts identities and differential equations) and how tools from mathematics (*twisted cohomology* and *intersection numbers*) can deepen their understanding. We focus on scalar Feynman integrals, *i.e.* those whose numerator is 1. All the loop integrals which appear in any perturbative Quantum Field Theory can be reduced to scalar Feynman integrals via a procedure called *Tensor reduction*.
The main source for this talk is S. Weinzierl's *Feynman Integrals* (open access), however it does also take formulae, explanations and examples from the first papers that introduced the connection between Feynman Integals and twisted cohomology, starting from *Feynman Integrals and intersection theory* by S. Mizera and P. Mastrolia. The last section is taken from *Cosmology meets cohomology*, A. Pokraka and S. De.

1.1 Momentum space representation

Let us define a generic scalar Feynman integral as

$$I_{\nu_1,\dots,\nu_{N_{int}}}(\{\underline{x}\}) = \int_{\mathbb{R}^{\mathbb{D}}} \prod_{r=1}^{L} d^D l_r \prod_{j=1}^{N_{int}} \frac{1}{(q_j(\{l\},\{x\})^2 - m_j^2)^{\nu_i}}, \tag{1}$$

where

- N_{int} is the number of internal lines;
- $D \in \mathbb{C}$ is the dimension of the integration;
- ν_i's are complex parameters;
- L is the number of loops;
- $\underline{x}$ is the set of external variables. These are scalar products of the $N_{ext} - 1$ independent external momenta and of the masses of the particles flowing both in the internal and external lines. Given the homogeneity properties of Feynman integrals, they can all be rescaled by a common quantity and taken to be dimensionless. Depending on how many massless particles there are this number can vary but it is bounded from above by

$$\frac{N_{ext}(N_{ext}-1)}{2} + N_{int} - 1; \tag{2}$$

- q_i's are linear functions of the external and loop momenta which are given by the topology of the graph. The external particles are also referred to as *legs*.

Based on their topology, a single Feynman integral can get its own name (bubble, triangle, box, double-box, sunrise (or sunset, or banana), caterpillars, train tracks...). A database of Feynman integrals featured in the literature is available at `http://loopedia.mpp.mpg.de/`.

1.2 Integration-by-parts identities

Let us now consider a fixed topology with a given assignment of kinematics. The integer values of the $\underline{\nu}$ vector defines a lattice of Feynman integrals referred as *family*. Notice that when some of the indices are non-positive the topology of the Feynman integral simplifies. Feynman integrals in the same family are related by *integration-by-parts* identities.

For any linear combination of loop and external momenta q_{IBP} and for any loop momentum l_i

$$\int \prod_{r=1}^{L} d^D l_r \frac{\partial}{\partial l_i^\mu} (q_{\text{IBP}}^\mu \mathcal{I}) = 0, \tag{3}$$

where $\mathcal{I}$ is a short form for the integrand.

Expanding the integrand gives the desired identities. Consider the equal-mass bubble diagram. This is a $L = 1$, $N_{ext} = 2$, $N_{int} = 2$ Feynman diagram depending on the physical quantities p^2 and m^2, i.e. by a single dimensionless variable that we choose to be $x = p^2/m^2$. The integral is

$$I_{\nu_1,\nu_2}(x) = \int d^D l \frac{1}{[(l-p)^2 - m^2]^{\nu_1}[l^2 - m^2]^{\nu_2}} = \int d^D l \frac{1}{[l^2 - 2(pl) + p^2 - m^2]^{\nu_1}[l^2 - m^2]^{\nu_2}}. \tag{4}$$

Choosing q^μ_{IBP} to be $-2p^\mu - xk^\mu$ and $2p^\mu - xk^\mu$ we get

$$\nu_1 x(4+x) I_{(\nu_1+1)\nu_2} = [2(-\nu_1+\nu_2) + (\nu_1 + 2\nu_2 - D)x] I_{\nu_1\nu_2} + \nu_1(2+x) I_{(\nu_1+1)(\nu_2-1)} - 2\nu_2 I_{(\nu_1-1)(\nu_2+1)}; \tag{5}$$

$$\nu_2 x(4+x) I_{\nu_1(\nu_2+1)} = [2(\nu_1-\nu_2) + (\nu_1 + 2\nu_2 - D)x] I_{\nu_1\nu_2} - 2\nu_1 I_{(\nu_1+1)(\nu_2-1)} - 2\nu_2 I_{(\nu_1-1)(\nu_2+1)}. \tag{6}$$

These identities can be used to reduce the calculation of any integral in the family to three *master integrals* I_{00}, I_{10} and I_{01}. The first retains the bubble topology, the other two are *tadpoles* integral. Using the symmetry of integration the master integrals can be reduced to only I_{00} and I_{10}

Integration-by-parts can get quite expensive computationally. The reduction procedure becomes very complicated and there are many choices on how to carry on the calculation. Several strategies have been developed to make the most common algorithm (Laporta algorithm) more efficient.

1.3 Differential equations

We have now reduced the problem of the computation of a generic Feynman integral to a finite set of master integrals. The modern approach to the computation of this integrals is via differential equation. The way it works is similar to Feynman's trick of integration: differentiate the integral with respect to an external parameter and set-up in this way a differential equation for the integral. In particular, it can be argued that by differentiating $I_{\underline{\nu}}(\{x\})$ with respect to an external parameter, gives a linear combination of integrals of the same family. Via integration-by-parts identities one can recast the R.H.S as a linear combination of master integral. Repeating this for every master integral gives a coupled system of linear differential equations

$$\frac{\partial}{\partial x_i} \underline{I}(\underline{x}) = A_i \underline{I}(\underline{x}), \tag{7}$$

where A_i is $N_{int} \times N_{int}$ matrix of rational functions.
For the equal-mass bubble the system turns out to be quite trivial

$$\frac{d}{dx}\begin{pmatrix} I_{11}(x) \\ I_{10}(x) \end{pmatrix} = \begin{pmatrix} -\frac{D-2}{x(4+x)} & -\frac{4+(4-D)x}{2x(4+x)} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} I_{11}(x) \\ I_{10}(x) \end{pmatrix}. \tag{8}$$

The solution can be found order-by-order by expanding around the regulator parameter $D = 4 - 2\varepsilon$.

In the multivariate case the differential equation can be elegantly repackaged in the language of differential forms. Defining

$$\mathrm{d} = \sum_{i=1}^{N_B} dx_i \frac{\partial}{\partial x_i}, \tag{9}$$

the equation becomes

$$d\underline{I}(\underline{x}) = A\underline{I}, \tag{10}$$

where A is a 1-from valued matrix.

1.4 Formalisation

A small detour. By construction, it can be shown that the matrix A satisfies the *flatness condition*

$$dA + A \wedge A = 0. \tag{11}$$

This suggests an interesting geometrical interpretation to the differential equation. A vector of master integral is an assignment of N_{master} complex numbers to a tuple in a complex projective space $(x_1 : \cdots : x_{N_B+1})$. A way to think of this is in terms of a vector bundle. The base space is the kinematic space $\mathbb{PC}^n$ and fibers are $\mathbb{C}^{N_{\text{master}}}$. The vector bundle is equipped with a connection

$$\nabla = d + A\wedge, \tag{12}$$

that satisfies the flatness condition and, thus, the restricted holonomy group of the vector bundle is the identity. This defines a local system and the connection is called Gauss-Manin connection. Master integrals are then flat section on a local system, *i.e* solutions to the parallel transport differential equation. The flatness condition implies that their solutions are iterated integrals on a contour that are well defined up to homotopy. Choice of different contours up to homotopy gives the multivalued-ness of the functions.

2 Twisted cohomology

2.1 Motivation: Aomoto Gel'fand hypergeomtric functions

We have found that IBP identities relate different Feynman integral in such a way that is possible to define what we have called a *basis*, whose size is N_{master}. It is natural to ask whether it is possible to define an actual *space* of integrals such that its dimension can be computed to be N_{master}. In order to do this, we need to study Feynman

integrals from a more abstract point of view. This can be achieved by moving to the *Baikov representation* of Feynman integrals, that is, essentially, a convenient change of integration variables. Under the Baikov representation a Feynman integral gets the general form

$$I(\underline{x}) = \int_C u(\underline{x}, \underline{z})\phi(\underline{x}, \underline{z}), \tag{13}$$

where

- u (the *twist*) is a multivalued function in the integration variables;
- ϕ is a n-holomorphic form;
- u vanishes on the boundary of C.

Integrals of this kind are called *Aomoto-Gel'fand hypergeometric functions* and constitute a multivariate generalisation of the classic hypergeometric functions. For example, the Baikov representation of the equal-mass bubble is

$$I_{\nu_1\nu_2}(x) = \int_\Gamma B(x, z_1, z_2)^{\frac{D-3}{2}} \frac{dz_1 \wedge dz_2}{z_1^{\nu_1} z_2^{\nu_2}}, \tag{14}$$

B is the *Baikov polynomial*

$$B(x, z_1, z_2) = -\frac{1}{4}(z_2 - z_1)^2 + \frac{1}{2}x(z_1 + z_2 + 2) - \frac{1}{4}x^2. \tag{15}$$

Since the exponential is generically complex the B^γ is multivalued[1] and, by construction of the Baikov representation, $B > 0$ on Γ and vanishes on its contour.

Now, the key insight here is that, having fixed u we study the the space of forms ϕ that, paired with the contour C, gives the integral

$$I = \langle \phi | C] = \int_C u\phi. \tag{16}$$

If $u = 1$ these integrals are compatible with the usual de Rham cohomology, that is

$$\text{if } \phi \sim \eta \text{ then } \langle \phi | C] = \langle \eta | C]. \tag{17}$$

Now with a generic twist we find that

$$\langle \phi | C] = \langle \eta | C] \text{ if } \eta = \phi + \nabla_\omega \xi, \tag{18}$$

where

$$\nabla_\omega \xi = \mathrm{d}\xi + \omega \wedge \xi, \tag{19}$$

where $\omega = \mathrm{d}\ln(u)$. The *de Rham twisted cohomology group* $H^n_\omega(M)$ is the space of closed formula modulo exact forms with respect to the covariant derivative $\nabla_\omega = d + \omega\wedge$. Then

The number of master integrals is the dimension of the twisted cohomology group.

(Here the I part of the seminar ends).

2.2 Review of de Rham Cohomology and Homology groups

A good reference for physicists to know more about this topic is Chapter 3 and 6 M. Nakahara's *Geometry, Topology and Physics* or (for those in a rush) Chapter 14 of S. Weinzierl's Feynman integrals.
Given a manifold M, let $\Omega^p(M)$ be the space of p-forms on M. If M is a n-dimensional manifold, $p = 0, \dots, n$. Recall that the differential d maps p-forms to $(p+1)$-forms. The p-th de Rham cohomology group is the equivalence class

$$H^p_{\mathrm{d}R}(M) = \{\eta \in \Omega^p(M), \mathrm{d}\eta = 0\}/\sim, \tag{20}$$

where

$$\eta \sim \eta' \quad \Longleftrightarrow \quad \exists\, \alpha \in \Omega^{p-1}(M), \eta = \eta' + \mathrm{d}\alpha. \tag{21}$$

In words, $H^p_{\mathrm{d}R}(M)$ is the space of closed forms modulo exact forms. $H^p_{\mathrm{d}R}(M)$ is a vector space ($0\mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_p}$ is a closed form).
A closely related concept is that of homology. Instead of forms one considers *p-chains*, that can intuitively thought as p-dimensional smooth submanifolds of M (paths for $p = 1$, surfaces for $p = 2$, etc...) On the space of $p-$chains on M, $\Omega_p(M)$, one can define the boundary map ∂ that maps $p-$chains to $(p-1)-$chains. Chains can be added and multiplied by scalars. The p-th de Rham homology group is the equivalence class

$$H_p(M) = \{\eta \in \Omega_p(M), \partial\eta = 0\}/\sim, \tag{22}$$

where

$$\eta \sim \eta' \quad \Longleftrightarrow \quad \exists\, \alpha \in \Omega^{p+1}(M), \eta = \eta' + \partial\alpha. \tag{23}$$

In words, $H_p(M)$ is the space of closed chains modulo a boundary. By the *Poincaré duality*, $H_p(M) \cong H^{n-p}_{\mathrm{d}R}(M)$. The dimensions of the groups are called *Betti numbers* and they give information on the topology of M (number of connected components, orientability, number of *holes*, the Euler characteristic...). Finally, we extend ordinary homology with *relative homology* $H_p(M, Y)$. In relative homology a closed chain admits a boundary inside Y. Closed chain are taken modulo boundaries or parts of Y (relative boundaries).

[1] Recall that in complex analysis a complex power is defined via $a^z = e^{z\log(a)}$, and log is a multivalued function.

2.3 Informal definition of twisted cohomology

We work on a generic complex $n-$dimensional manifold A[2]. Let $\underline{z} = (z_1, \dots, z_n)$ be its holomorphic coordinates in a given chart $U \subset \mathbb{C}^n$, and take m irreducible polynomials $\{p_1, \dots, p_m\}$. In the following, we will always work on a chart and ignore the global structure of M. The hypersurface defined by the vanishing of their locus is called a *divisor*,

$$D = \{\underline{z} \in U, \exists p_i(\underline{z}) = 0\}. \tag{24}$$

A generic rational holomorphic n-form on $M = A\backslash D$ is

$$\phi(\underline{z}) = \frac{q(\underline{z})}{p_1^{n_1}, \dots, p_m^{n_m}} \mathrm{d}z_n \wedge \dots \wedge \mathrm{d}z_1, \tag{25}$$

where q is a polynomial and n_i's are integers. The form is holomorphic because it depends only on the holomorphic coordinates. Since there are exactly n of such coordinates, by antisymmetry of the wedge product there is no $(n+1)$-holomorphic form and ϕ is said to be a *top holomorphic form*.

Now, define the *twist* function $u = \prod_{i=1}^m p_i^{\gamma_i}$, with γ_i complex parameters and $\omega = \mathrm{d}\ln u$. The *twisted cohomology group* on M is the de Rham cohomology group where the differential d is replaced with the covariant derivative $\nabla_\omega = d + \omega\wedge$:

$$H^p_\omega(M) = \{\eta \in \Omega^p(M), \nabla_\omega \eta = 0\}/\sim_\omega, \tag{26}$$

where

$$\eta \sim_\omega \eta' \quad \iff \quad \exists\, \alpha \in \Omega^{p-1}(M), \eta = \eta' + \nabla_\omega \alpha. \tag{27}$$

Top holomorphic forms (**??**) are clearly in H^n_ω because they are closed, $\mathrm{d}\phi + \omega \wedge \phi = 0$ because both summands are $(n+1)$-holomorphic forms on a n-dimensional complex space. Since $H^n_\omega(M)$ is a vector space, it is possible to find

1. its dimension;
2. a basis;
3. an inner product.

After a few examples, in the following sections we will quickly go through each of these items. These will make possible to find identities between functions which, in case of Feynman integrals, recover integration-by-parts identities.

2.4 Back to Aomoto-Gel'fand hypergeometric functions

Now we can start to put together all the pieces. Having fixed a twist u, an Aomoto-Gel'fand hypergeometric function is defined as a pairing of a top holomorphic form ϕ in $H^n_\omega(A\backslash D)$ and a chain C in the relative homology group $H_n(A, D)$,

$$I(\underline{x}) = \langle\phi|C] = \int_C u(\underline{x}, \underline{z})\phi(\underline{x}, \underline{z}). \tag{28}$$

$\langle\phi|$ and $|C]$ emphasise that the form and the chain are defined in an equivalence class, they both can be deformed by exact forms or relative boundaries and the integral remains the same. Let us consider a one-dimensional example

$$I(\{x_i\}) = \int_0^1 z^{\gamma_1}(1-z)^{\gamma_2} \prod_{i=3}^{a} (1 - x_{i-2}z)^{\gamma_i}. \tag{29}$$

For $a = 2$ we find the the Beta function $\beta(\gamma_1, \gamma_2)$, for $a = 3$ the integral reduces to classic Gauß hypergeometric function ${}_2F_1$ and for $a = 4$ $I(x_1, x_2)$ is Appel's function F_2, a generalisation of ${}_2F_1$. The twist is the whole integrand and $\phi = 1\mathrm{d}z$. The underlining space is the punctured projective line $\mathbb{C}\backslash\{0, 1, x_1^{-1}, \dots, x_{a-3}^{-1}\}$. Let us focus on $a = 2$.

Gauß hupergeometric function is defined[3] as

$${}_2F_1(a, b; c; x) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(b-c)} \int_0^1 z^{b-1}(1-z)^{c-b-1}(1-xz)^{-a}\mathrm{d}z. \tag{30}$$

Let us write it as

$$\langle\phi|C] = \int_0^1 z^b(1-z)^{c-b}(1-xz)^{-a} \frac{\mathrm{d}z}{z(1-z)}, \tag{31}$$

$\phi = \frac{\mathrm{d}z}{z(1-z)}$ is a holomorphic 1-form on $\mathbb{C}\backslash\{0, 1, \frac{1}{x}\}$ (a space with 2 real dimensions and 1 complex dimension). The twist form is

$$\omega = \mathrm{d}\ln u = b\frac{\mathrm{d}z}{z} + (c-b)\frac{\mathrm{d}z}{1-z} - a\frac{\mathrm{d}z}{1-xz}. \tag{32}$$

As argued in the previous section, $\nabla_\omega \phi = 0$ and hence ϕ is a representative form of an equivalence class in $H^1_\omega(\mathbb{C}\backslash\{0, 1, x^{-1}\})$. Gauß hypergeometric satisfies many identities. Examples are Gauss contiguous relations, such as

$${}_2F_1(a, b; c+1; x) = \frac{c}{c-a}\, {}_2F_1(a, b; c, x) + \frac{a}{a-c}\, {}_2F_1(a+1, b; c+1; z). \tag{33}$$

These can be translated into an identity of forms in twisted cohomology as

[2]For a manifold to be complex of dimension n, it means that it is a real manifold of dimension $2n$ such that for every chart with coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ there exists a rearrangement - compatible with all other charts - of the coordinates into n *holomorphic* and n *anti-holomorphic* coordinates $(z_k = x_k + iy_k, \bar{z}_k = x_k - iy_k)$. Not all even-dimensional manifolds admit such a *complex structure*.

[3]This is the integral representation. The function was first defined as a power series generalising the geometric series, hence the name. Equivalently, ${}_2F_1$ can also be defined via differential equations.

$$\langle\phi| = \frac{c}{c-a}\langle\phi_1| + \frac{a}{a-c}\langle\phi_2|, \tag{34}$$

where, form the integral representation of ${}_2F_1$, we have

$$\begin{aligned}\phi &= \frac{c}{c-b}\frac{\mathrm{d}z}{z},\\ \phi_1 &= \frac{\mathrm{d}z}{z(1-z)},\\ \phi_2 &= \frac{c}{c-b}\frac{\mathrm{d}z}{(1-xz)z}.\end{aligned} \tag{35}$$

while it would be quite cumbersome to generate this kind of identities from the integral representation, twisted cohomology gives a tool to construct them algorithmically. In practice, equation (**??**) should be interpreted as decomposing the form ϕ on a basis of form in the twisted cohomology group. The coefficients, then, are just projections obtained by computing an inner product on the vector space $H^1_\omega(\mathbb{C}\backslash\{0,1,x^{-1}\})$. This is also the way integration-by-parts identities are interpreted in this framework.

2.5 $H^n_\omega(M)$ as a vector space

Dimension The following facts on the dimension of $H^n_\omega(M)$ can be proved under certain assumptions

(i) $\dim H^n_\omega(M) = \dim H_n(A,D)$;

(ii) The only cohomology group $H^p_\omega(M)$ with dimension higher than 0 (non-trivial group) is the middle group, *i.e* $p=n$;

(iii) From the previous fact we find $\dim H^n_\omega(M) = |\chi(M)|$, where χ is the Euler characteristic;

(iv) Morse theory (or its complex version, Picard-Lefschetz theory), allows to compute $\chi(M)$ as the number of critical points of u.

Putting the last three facts together we find

$$\dim H^n_\omega(M) = \text{number of solutions to } \omega(\underline{z}) = 0, \tag{36}$$

where we are taking generic values of $\underline{x}$ such that the zeroes of ω are all isolated and in M. In the case of ${}_2F_1$ we have

$$\omega = \frac{\text{II degree equation}}{z(1-z)(1-xz)}\mathrm{d}z, \tag{37}$$

and therefore

$$\dim H^1_\omega(\mathbb{C}\backslash\{0,1,x^{-1}\}) = 2. \tag{38}$$

The first fact allows for an alternative computation of the dimension in the case where M is a complement of an *hyperplane arrangement*, that is, the polynomials p are all linear. In that case (under certain assumptions) the dimension of $\dim H_n(A,D)$ is the number of bounded regions in the *real part* of A. In practice, in the case of $\mathbb{C}\backslash\{0,1,x^{-1}\}$, the real part is $\mathbb{R}\backslash\{0,1,\Re[x^{-1}]\}$, which has clearly two bounded regions, so that we retrieve the correct dimension.

Basis We limit ourselves to just mention three ways to find a basis of the twisted cohomology.

- The monomial basis via Groebner basis;
- dlog basis via positive geometry;
- A basis via Lefschetz thimbles.

In the first case one lists out all the monomials in $\underline{z}$ in lexicographic order and filters out some based on divisibility by elements in the Groebner basis of the ideal generated by $\{p_1,\dots,p_m\}$. The second can only be applied in the case of the complement of hyperplanes: basis is given by the canonical forms (in the sense of positive geometry) of the bounded region. Such forms can always be written as dlog's.

Inner product The inner product on $H^n_\omega(M)$ is called *intersection number*, as it is related to the inner product on the homology (where one can think, geometrically, as intersection of chains). We leave the formal definition aside, but simply mention that, while the calculation for 1-dimensional forms is relatively straightforward, the computation of a generic intersection number can become quite complicated, even though an iterative algorithm on the dimension exists. In the case of dlog forms a closed formula exists for any dimension. Having determined a basis for $H^n_\omega(M)$, both the integration-by-part reduction and the determination of a differential equation is reduced to the problem of computing intersection numbers.

3 An example from Cosmology

The *wavefunction coefficient* in a power-law FLRW spacetime for a conformally-coupled scalar with generic polynomial self-interaction reads

$$\psi_2(X_1,X_2,Y_{12}) = \int_0^\infty\int_0^\infty (x_1)^{\alpha_1}(x_2)^{\alpha_2}\frac{\mathrm{d}x_1\mathrm{d}x_2}{(X_1+Y_{1,2}+x_1)(X_2+Y_{1,2}+x_2)(X_1+X_2+x_1+x_2)}. \tag{39}$$

The integral is already in a suitable form for twisted cohomology, so it does need any further change of variable as the Baikov representation for Feynman Integrals. However, as it is often the case for applications, the twist is not,

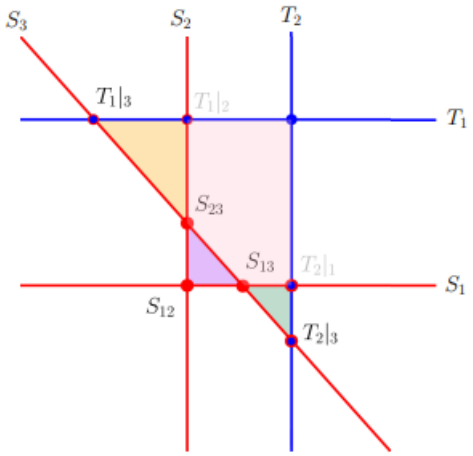


Figure 1: For generic values of the external kinematic X_1, X_2 and $Y_{1,2}$, the 5 divisors define four bounded regions. Figure from *Cosmology meets cohomology*, A. Pokraka, S. De,arXiv:2308.03753.

naıvely, $u = x_1^{\alpha_1} x_2^{\alpha_2}$, but, since the 2-form has three extra singularities, u should include also the other three divisors with arbitrary exponents. Then, one can see (Figure **??**) that $\omega = \mathrm{dln} u$ has 4 zeroes and thus

$$\mathrm{dim} H^2_\omega(\mathbb{C}^2 \backslash \{x_1 = 0, x_2 = 0, X_1 + Y_{1,2} + x_1 = 0, X_2 + Y_{1,2} + x_2 = 0, X_1 + X_2 + x_1 + x_2 = 0\}) = 4. \tag{40}$$

Since the divisors are all linear, the underlying space is a complement of hyperplane arrangements. As stated in the previous section, the dimension can also be found by counting the number of bounded regions in the real slice of the space and the basis can be constructed from the *canonical forms* of the four regions. Since canonical forms are dlog forms, the intersection numbers are straightforward to compute.